

MATH 114 HW 2

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Question 1

Define the sample mean $\bar{X}_n$ and sample variance S_n^2 as

$$\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k \quad S_n^2 = \frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X}_n)^2$$

Recall $\sigma^2 = \mathbb{E}(X^2) - \mu^2 \Rightarrow \sigma^2 - \mu^2 = \mathbb{E}(X^2)$

$$\begin{aligned} S_n^2 &= \frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X}_n)^2 \\ &= \frac{1}{n-1} \left(\sum_{k=1}^n X_k^2 - n\bar{X}_n^2 \right) \\ \mathbb{E}(S_n^2) &= \frac{1}{n-1} \left(\sum_{k=1}^n \mathbb{E}(X_k^2) - n\mathbb{E}(\bar{X}_n^2) \right) \\ &= \frac{1}{n-1} \left(\sum_{k=1}^n (\mu^2 - \sigma^2) - n((\mathbb{E}(\bar{X}))^2 + \text{Var}\bar{X}) \right) \\ &= \frac{1}{n-1} \left(n\mu^2 + n\sigma^2 - n(\mathbb{E}(\frac{1}{n} \sum_{k=1}^n X_k)^2 + \text{Var}(\frac{1}{n} \sum_{k=1}^n X_k)) \right) \\ &= \frac{1}{n-1} \left(n\mu^2 + n\sigma^2 - n((\frac{1}{n} \sum_{k=1}^n \mathbb{E}(X_k))^2 + (\frac{1}{n^2} \sum_{k=1}^n \text{Var}(X_k))) \right) \\ &= \frac{1}{n-1} \left(n\mu^2 + n\sigma^2 - n((\frac{1}{n} \cdot n\mu)^2 + \frac{1}{n^2} \cdot n\sigma^2) \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n-1} (n\mu^2 + n\sigma^2 - n\mu^2 - \sigma^2) \\
&= \frac{1}{n-1} (n-1)(\sigma^2) \\
&= \sigma^2
\end{aligned}$$

Question 2

Define $I = \mathbb{E}(g(X))$ and

$$I = \int_D g(x)f(x)dx \qquad I_n = \frac{1}{n} \sum_{k=1}^n g(X_k)$$

We have

$$\begin{aligned}
\text{Var}(I_n) &= \text{Var}\left(\frac{1}{n} \sum_{k=1}^n g(X_k)\right) \\
&= \frac{1}{n^2} \sum_{k=1}^n \text{Var}(g(X_k)) \\
&= \frac{1}{n} \text{Var}(g(X)) \\
&= \frac{1}{n} (\mathbb{E}(g(X)^2) - \mathbb{E}(g(X))^2) \\
&= \frac{1}{n} \left(\int_D g(x)^2 f(x) dx - I^2 \right)
\end{aligned}$$

Question 3

N	I_N	$\text{Var}(I_N)$	J_N	$\text{Var}(J_N)$
100	0.8107	0.0026	0.8276	0.0017
400	0.8177	0.00086	0.8527	0.00036
900	0.8423	0.00029	0.8266	0.00020
1600	0.8174	0.00017	0.8397	0.000079

Code for the simulation is included below.

```
import numpy as np
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N = 100
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```
def crude_monte_carlo_integration():
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X_n = np.random.uniform(0, 4, N)
sum = 0
for x in X_n:
    sum += 4 * (np.sqrt(x) * np.exp(-x))
I = 1 / N * sum
return I

def importance_sampling_integration():
    U = np.random.uniform(0, 1, N)
    Y_n = -np.log(1 - U * (1 - np.exp(-4)))

    sum = 0
    for y in Y_n:
        sum += (np.sqrt(y) * np.exp(-y)) / (np.exp(-y) / (1 - np.exp(-4)))
    I = 1 / N * sum
    return I

def samp_var(X):
    sum = 0
    for i in range(len(X)):
        sum += (X[i] - np.mean(X)) ** 2
    return sum / (len(X) - 1)

results_mc = []
for i in range(1, 101):
    results_mc.append(crude_monte_carlo_integration())
var_mc = samp_var(results_mc)
crude_monte_carlo_integration(), var_mc

results_is = []
for i in range(1, 101):
    results_is.append(importance_sampling_integration())
var_is = samp_var(results_is)
importance_sampling_integration(), var_is

```

Question 4